

Innovation Experiment - Significance Statement

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NIH #1: Importance of the Problem

The study of drug resistance is implicated in the majority of cancer related deaths in patients with metastatic disease [1, 2]. Even though we have targeted therapies that are designed to block specific oncogenic drivers, most patients acquire resistance within one or two years of the initiation of treatments [2]. The issue today is not the shortage of data. We have huge public repositories like GDSC2, CCLE and CTRPv2 that contain pharmacological sensitivity profiles mapped with genomic, transcriptomic and pathway annotations for several cell line-drug pairs. Recent work even shows that general principles of drug sensitivity are completely learnable from these repositories [3]. What we do not see today is a good framework that builds reason over all the biological layers, analogous to how oncologists do today. It requires a causal ordering of the evidence gathered and if any of the layers disagree, this causal ordering should decide the outcome, and not the statistical average.

NIH #2: Scientific Premise

We hypothesize that a decentralized multi-agent framework resolving inter-modality conflicts through an explicit 5-tier hierarchy of truth will predict the drug resistance more accurately and interpretably in comparison to monolithic LLMs and naive systems relying on majority-voting.

Three lines of evidence support this.

First, we already have this causal hierarchy defined in biology. Structural DNA alterations are upstream of transcriptional state and this in turn is upstream of protein activity. This consequently determines the pathway flux [1, 3]. The core novelty lies in the fact that we are modeling this as an executable logic rather than leaving it as model weights.

Second, Four specialized agents (genomics, transcriptomics, pharmacology and pathways) query a curated local database using Model Context Protocol [11, 12] and output a structured EvidencePack. This is input into the DebateEngine that runs a maximum of 3 rounds of conflict resolution using a ConflictDetector and AxiomResolver. The 5 tiers of hierarchy are as follows:

- **T1 - Structural Primacy:** DNA-level loss-of-function overrides all downstream signals.
- **T2 - Transcriptional Gate:** Silencing (z-score < -2.0) renders a gene functionally absent.

- **T3 - Pathway Bypass:** Active bypass routes confer resistance regardless of target status.
- **T4 - Pharmacological Prior:** Historical IC_{50} is a population baseline, overridable by T1–T3.
- **T5 - Statistical Consensus:** Population statistics apply when molecular evidence is absent.

The output of every prediction will also include a structured debate trace which will mention the name of the axiom tier that helped resolve the case, which agents’ evidences were overridden and which of the dissents were preserved. This is a form of clinical accountability that current AI systems do not provide [6, 7, 8].

Third, we see some weaknesses in prior work that can be addressed. Classical ML models are black-boxes and their predictions do not include any transparency on the mechanism used to inspect which data layer drove the outcome [4, 5]. LLMs suffer from context overwhelm when disjoint data types are presented altogether. They also suffer from hallucination which is a well-documented clinical risk [9, 10]. We intend to address both of these issues in our study by anchoring every claim to a database query and every conflict among agents is resolved by a logged axiom logic.

NIH #3: Improvement to Scientific Knowledge and Clinical Practice

Aim 1 establishes predictive accuracy on the development set from GDSC2 (AUROC, AUPRC, Cohen’s κ , Spearman’s ρ). **Aim 2** quantifies the quality of reasoning using hallucination rate and biological faithfulness score. These are dimensions that were not measured in any previous work [9]. **Aim 3** is to run 6 ablation studies (no axiom hierarchy, single agent, free-text data instead of MCP, randomly ordering tiers, no debate loop, majority vote) to show that each of the architectural choices made are independently valid.

Scientific benefit: We produce a reproducible, LLM-agnostic multi-agent framework capable of biological reasoning.

Technical benefit: An open-source MCP Agent-to-Tool protocol that can be generalized to any biomedical task [13].

Clinical benefit: Providing structured reasoning traces that can be used by clinicians to interrogate scientifically.

NIH #4: How the Field Will Change

If these aims are achieved, then we can observe a shift in the optimization target from AUROC points towards reasoning quality metrics. We can generalize the axiom-governed debate protocol to any domain that requires multi-modal evidence reconciliation (like rare

disease diagnosis, treatment response prediction, biomarker discovery [13]). Moreover, the debate trace is in itself a novel artifact. It is a citable, structured record showing the evidence tier that resolved the case which enables clinicians to monitor the whole debate process and not just the outcome of the debate [6, 7].

Consequently, our knowledge of how multi-modal systems should be integrated for clinical inference will be aided by an auditable and biologically faithful architecture. This can be the first step to bring computational oncology one step closer to AI-assisted decision support that clinicians can trust.

Reflection

The Wrenshall sock example helped me understand how to structure the hypothesis and put preliminary data together before chalking out a plan. It is crucial to believe in the solution before we care about how to execute a plan to achieve the solution. This experiment has helped me to take a step back from the technicality of the process and see what the problem is to the outside world and how it all fits into a bigger picture. It has also taught me to assert the scope of my problem using literature and data to make a strong claim on why a new logic for this problem statement is needed.

Feedback

One of the crucial feedbacks given by my project guide (Dr. Bhaskarjyoti Das) is to include data and past literature comments as much as possible to make our architecture look more achievable than it just being a hypothetical design. This feedback has directly helped in this experiment for me to structure the way in which I present my problem statement as an Innovation Experiment that has the capability to influence how the field of computational oncology views AI-assisted decision support systems.

References

- [1] Liu X et al. Multi-omics advances in understanding cancer drug resistance. *Biomed Technol.* 2025. doi:10.1016/j.bmt.2025.100119
- [2] Mao Y et al. Emerging AI-driven precision therapies in tumor drug resistance. *Mol Cancer.* 2025. doi:10.1186/s12943-025-02321-x
- [3] Carli F et al. Learning and actioning general principles of cancer cell drug sensitivity. *Nat Commun.* 2025. doi:10.1038/s41467-025-56827-5
- [4] Saranya KR, Vimina ER. DRN-CDR: cancer drug response prediction using multi-omics. *Comput Biol Chem.* 2024. doi:10.1016/j.compbiolchem.2024.108175
- [5] Zhao H et al. MSDRP: deep learning model for drug response prediction. *Bioinformatics.* 2023.

- [6] Liu Q et al. EvoMDT: self-evolving multi-agent system for clinical decision-making. *npj Digit Med*. 2026. doi:10.1038/s41746-025-02304-8
- [7] Han X et al. Multi-Agent Medical Decision Consensus Matrix for Oncology MDT. *arXiv:2512.14321*. 2024.
- [8] Lorenzo L et al. Evaluation of Oncotimia: LLM system for tumour boards. *arXiv:2601.19899*. 2026.
- [9] Kim Y et al. Medical Hallucinations in Foundation Models. *arXiv:2503.05777*. 2025.
- [10] Nishisako S et al. Reducing Hallucinations in Generative AI for Cancer Information. *JMIR Cancer*. 2025. doi:10.2196/70176
- [11] Flotho M et al. MCPmed: MCP-enabled bioinformatics for LLM-driven discovery. *Brief Bioinform*. 2026. doi:10.1093/bib/bbag076
- [12] Widjaja F et al. BioinfoMCP: MCP Interfaces in Agentic Bioinformatics. *arXiv:2510.02139*. 2025.
- [13] Mehandru N et al. BioAgents: multi-agent systems for bioinformatics. *Sci Rep*. 2025. doi:10.1038/s41598-025-25919-z